File # _____

CONTRACTOR'S DAILY ROCK BOLT TESTING AND TENSIONING RECORD

PROJECT NO. _____

(TO BE SUBMITTED WITHIN 1 DAY AFTER EACH DAY'S ROCK BOLTING OPERATION)

Contractor Name _____ Sample calculation for this grout: Date _____

Bolt Type _____ Size _____ Weight of each bag of grout (M) _____ kg Weather _____

Resin/Grout Type _____ Bond Zone Length ____ m Volume of water added for each bag (V) _____ L Temperature _____ °C

Test Jack Number(s) _____ Pump ID _____ Actual Water/Cement Ration (V/M) _____

Bond Zone Type: (circle) RESIN GROUT

Free Stressing Length Type: (circle) RESIN GROUT

Name of Person who Installed and Tested the Rock Bolt	Zone	Area	Bolt Sequence Number for the Area	Bolt Length (m)	Bolt Rotation Time Through Resin Start	Bolt Rotation Time Through Resin End	Test Start Time	Test Details: Record Gauge Reading of Each Minute Elapsed Time (minute) 1	2	3	4	5	6	7	8	9	10	Lock Off Load (kN)	Test End Time	Grout Cube #	Certified Correct by Tester (Signature)

Certified Correct: _____
(Contractor's Superintendent)

ROCK BOLTS — SP206–01

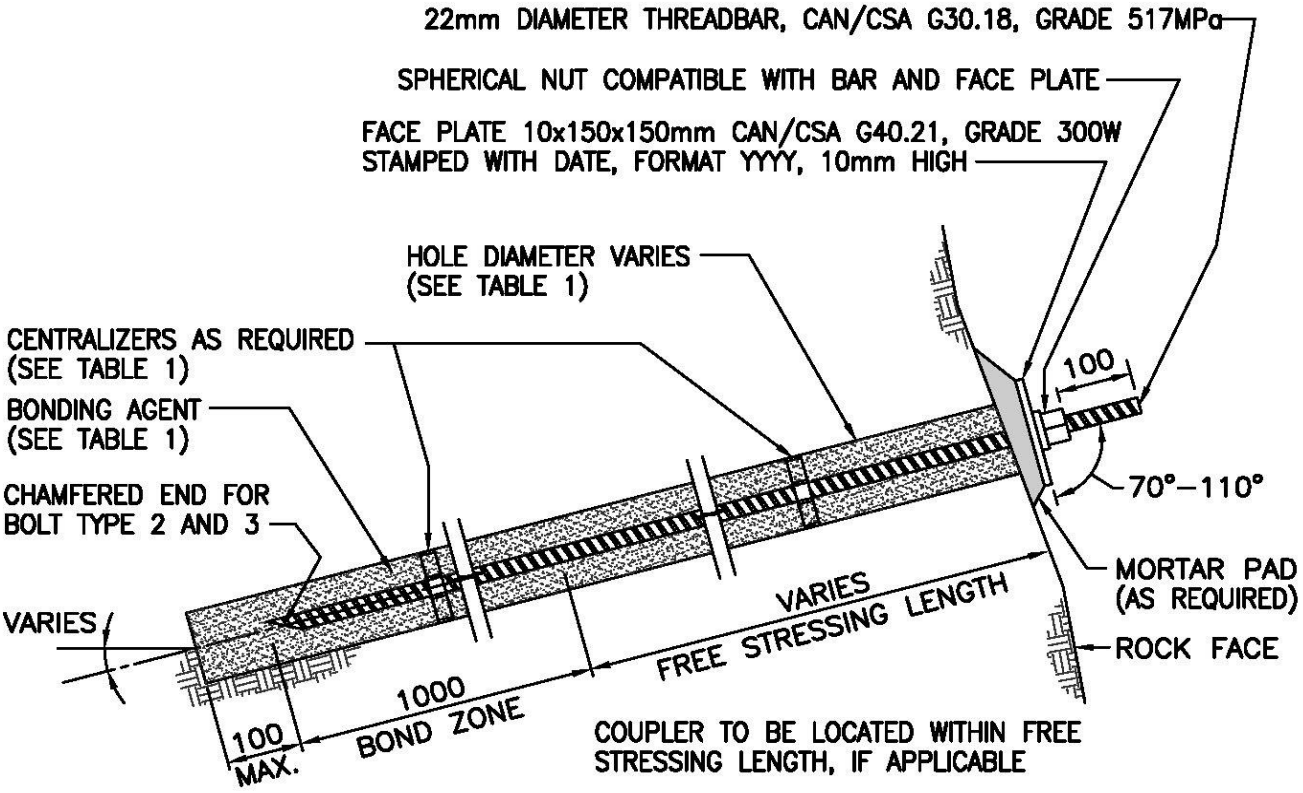


TABLE 1 BOLT TYPE, BONDING AGENT, HOLE DIAMETER AND CENTRALIZERS

BOLT TYPE *	BONDING AGENT		HOLE DIAMETER mm **		CENTRALIZERS
	BOND ZONE	FREE STRESSING LENGTH	BOND ZONE	FREE STRESSING LENGTH	
1	GROUT	GROUT	42 MIN.	42 MIN.	YES
2	RESIN	GROUT	38	42	NO
3	RESIN	RESIN	38	41	NO

* BOLT TYPE 3 MAY BE USED DUE TO FIELD CONDITIONS IF TYPE 1 AND 2 ARE NOT POSSIBLE AND ONLY IF AUTHORIZED BY THE MINISTRY REPRESENTATIVE. COUPLER IS NOT PERMITTED FOR BOLT TYPE 3.

** HOLE DIAMETER MAY VARY DEPENDING ON MATERIAL SUPPLIER. INCREASE HOLE DIAMETER IN FREE STRESSING LENGTH TO 56mm IF BAR COUPLER IS USED.

NOTES:

1. GROUT TUBE OMITTED FOR CLARITY.
2. TEST TENSION TO 186kN, HOLD FOR 10 MINUTES.
3. LOCK–OFF LOAD IS 140kN.
4. ALL STEEL TO BE HOT DIP GALVANIZED TO ASTM A123 OR A153, AS APPROPRIATE.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

SECTION 207
SLOPE MESH FOR ROCK CUTS

DESCRIPTION

207.01 Scope – This Section applies to slope mesh structures which are installed to provide rockfall protection.

207.01.01 Scaling – Rock slope scaling, tree removal and overburden excavation performed in conjunction with slope mesh installation shall be considered incidental to slope meshing. Rock slope scaling, tree removal and overburden excavation shall be completed before the mesh support system is installed unless otherwise authorized by the Ministry Representative. Trees to be removed will be detailed by the Ministry Representative.

207.01.02 Materials Disposal – Discarded mesh components, scaled rock, trees and debris generated by the slope meshing work shall be removed from the Site and disposed of by the Contractor in accordance with SS 145.27.02. Temporary storage of felled trees in ditches will not be permitted.

207.01.03 General Layout – The Contractor shall layout in the field all anchor and suspension cable locations for each section for review by the Ministry Representative prior to installation.

207.01.04 Layout Variations – Field conditions may require final anchor, cable and mesh configurations to vary from the Ministry's Drawings. All variations shall be authorized by the Ministry Representative.

207.01.05 Mesh Sections – The area to be provided with slope mesh protection shall be divided into sections with a maximum width of 24 m. Each section shall have a separate suspension cable, end main anchors and end anchors. End main anchors of adjacent sections shall be positioned 200 mm apart. The gap between sections shall be closed with mesh.

207.01.06 Mesh Height – The maximum mesh height shall be 80 m.

207.01.07 Impact Energy – The maximum rock impact energies shall be as follows:

(a) 3 kJ above cut crest; and

(b) 30 kJ below cut crest.

MATERIALS

207.11 Slope Meshing Materials – Unless otherwise specified, preparation and installation of materials shall be according to manufacturer's recommendations.

All components shall be hot dip galvanized by the manufacturer conforming to ASTM A123 or ASTM A153, as appropriate. Damaged galvanizing shall be re-galvanized at the Contractor's expense.

207.11.01 Mesh – Mesh shall be 11 gauge (2.95 mm dia.) hexagonal double-twist (i.e. three one-half turns) gabion type mesh. Mesh wire shall meet ASTM A975 style 1 consisting of carbon steel with a class 3 zinc coating of not less than 244 g/m² meeting ASTM A641. The weight of zinc coating shall be determined by ASTM A90. The coating shall withstand four one-minute dips by the Preece test, ASTM A239. Mesh opening shall be hexagonal in shape and uniform in size measuring 80 mm by 100 mm.

207.11.02 Threadbar – Threadbar shall be steel hot-rolled Grade 517 MPa, conforming to CSA G30.18.

207.11.03 Eye nuts – Eye nuts shall be cast or manufactured eye nuts supplied by the threadbar manufacturer.

207.11.04 Cables – All cables shall be fibre core conforming to CSA G4. Cables shall be unspliced.

207.11.05 Thimbles – Thimbles shall meet US Federal Specification FF-T-276 Type III.

207.11.06 Clips – Clips shall meet US Federal Specification FF-C-450 Type 1, Class 1.

207.12 Grout

207.12.01 Anchors – Main anchors shall be secured in the drill hole using grout. Auxiliary anchors shall be secured in the drill hole using grout or resin.

207.12.02 Grout – Grout shall have the following properties:

(a) be pre-mixed, unsanded, non-metallic, and non-shrink cementitious grout containing silica fume.

(b) may be mixed to a flowable consistency, typically with a water to cement ratio of 0.35 or as specified by the grout manufacturer to achieve or exceed the required strength.

(c) minimum 7 day compressive strength of 30 MPa and a minimum 28 day compressive strength of 40 MPa, tested in accordance with CSA A23.2-09.

(d) admixtures to be used according to the manufacturer's specifications and as authorized by the Ministry Representative.

(e) calcium chloride-based accelerators are not permitted.

207.12.03 Resin – Resin shall have the following properties:

(a) be supplied in cartridge form.

(b) a minimum unconfined compressive strength of 90 MPa when fully mixed and cured and tested in accordance with CSA A23.2-09.

(c) be encased in a plastic film that provides optimum resistance to moisture and is easily ruptured to enable complete mixing during installation.

(d) suitable thixotropic and viscous properties to permit adequate mixing of the resin components by rotation of the rock bolt bar and to contain the resin within the drill hole.

(e) an easily identifiable gel time recommended by the resin manufacturer.

(f) reach 80% of its ultimate strength within a time interval equal to five times the gel time.

(g) be non-shrink after the gel time.

(h) be unaffected by mild acids or mild alkalis.

(i) be in cartridge boxes labelled with the resin expiry date.

207.13 Conformance Documents – Prior to installation the contractor shall supply documents of conformance to project specifications of all materials upon request.

CONSTRUCTION

207.31 Anchor Installation

207.31.01 Overburden – Overburden shall be excavated to rock at anchor locations unless specified otherwise. The Contractor shall minimize disturbance of surrounding soil and rock when excavating. Overburden shall be removed as necessary so that cables do not contact ground surface.

207.31.02 Ground Clearance – The minimum ground clearance for the suspension cable shall be 0.6 m between anchors, or up to 1.5 m if so specified. A maximum of one connector may be used above the ground to extend any main anchor as necessary to achieve the required clearance. Suspension cable clearance shall be maximized where possible by appropriate selection of anchor locations.

207.31.03 All Anchors – Anchor holes shall be a minimum of 1.5 times anchor diameter and in strong, competent rock. Anchors shall be centered in the hole and grouted. Anchors shall not be loaded within 3 days of grouting.

207.31.04 Main Anchors – Main anchors shall be located at local high points where practicable to maximize clearance between suspension cable and ground surface. Main anchors shall be vertical and centered in the hole unless otherwise authorized by the Ministry Representative. Main anchor height above ground surface may be reduced where authorized by the Ministry Representative for field conditions.

207.31.05 Auxiliary Anchors – Auxiliary anchors shall be located to minimize potential for main anchor bending.

207.31.06 Guy Anchors and Cables – Main anchors may be substituted, where authorized by the Ministry Representative, with a limited number of guy cables directly connected to the suspension cable where no suitable main anchor locations can be found. Guy cable anchors shall be located to maximize suspension cable elevation. See Section Y-Y, on SS Drawing SP207-02 and Detail B on SS Drawing SP207-03.

207.31.07 Embedded Length – Auxiliary or guy anchor embedded length may require extension if weak rock conditions are encountered.

207.32 Soil Anchors

207.32.01 Soil Anchor Placement – Soil anchors are incidental to slope meshing and may be required in place of end main, main, guy and/or auxiliary anchors where:

(a) Bedrock is not at ground surface.

(b) Overburden thickness is greater than 0.6 m.

(c) Required at any other locations or as ordered by the Ministry Representative.

For soil anchor details see SS Drawings SP207-04 and SP207-05.

207.32.02 Soil Anchor Concrete Requirements – Concrete requirements for soil anchors:

(a) Minimum compressive strength at 28 days = 30 MPa

(b) Maximum nominal size of aggregate = 28 mm

(c) Air content = 5 ± 1%

(d) Slump = 55 ± 20 mm

(e) Maximum w/c ratio by mass = 0.45

207.32.03 Soil Anchor Load Testing – Upon request by the Ministry Representative, the Contractor shall load test 10% of the overburden soil anchors to 10 kN to verify capacity or as authorized by the Ministry Representative.

207.33 Cable and Mesh Installation – Suspension, auxiliary, end, and guy cables shall be installed to nominal tension to remove slack before and after installing mesh. Terminate lacing cable ends with a loop through the end main anchor eye nut, secured to the suspension cable or the lacing cable itself using two clips, 19 mm or 6 mm respectively.

A maximum of two horizontal mesh seams (200 mm overlap) shall be permitted along the entire mesh height. The upper mesh portion shall be between the slope and lower mesh at the overlap. The horizontal seam connections shall be similar to the vertical seams.

Unless otherwise specified on the Contract Drawings or by the Ministry Representative, the mesh shall terminate 1500 mm above the highway edge of pavement elevation. The bottom of the mesh shall be evenly trimmed parallel with the highway elevation. The bottom of the mesh shall be bent to remove the curl.

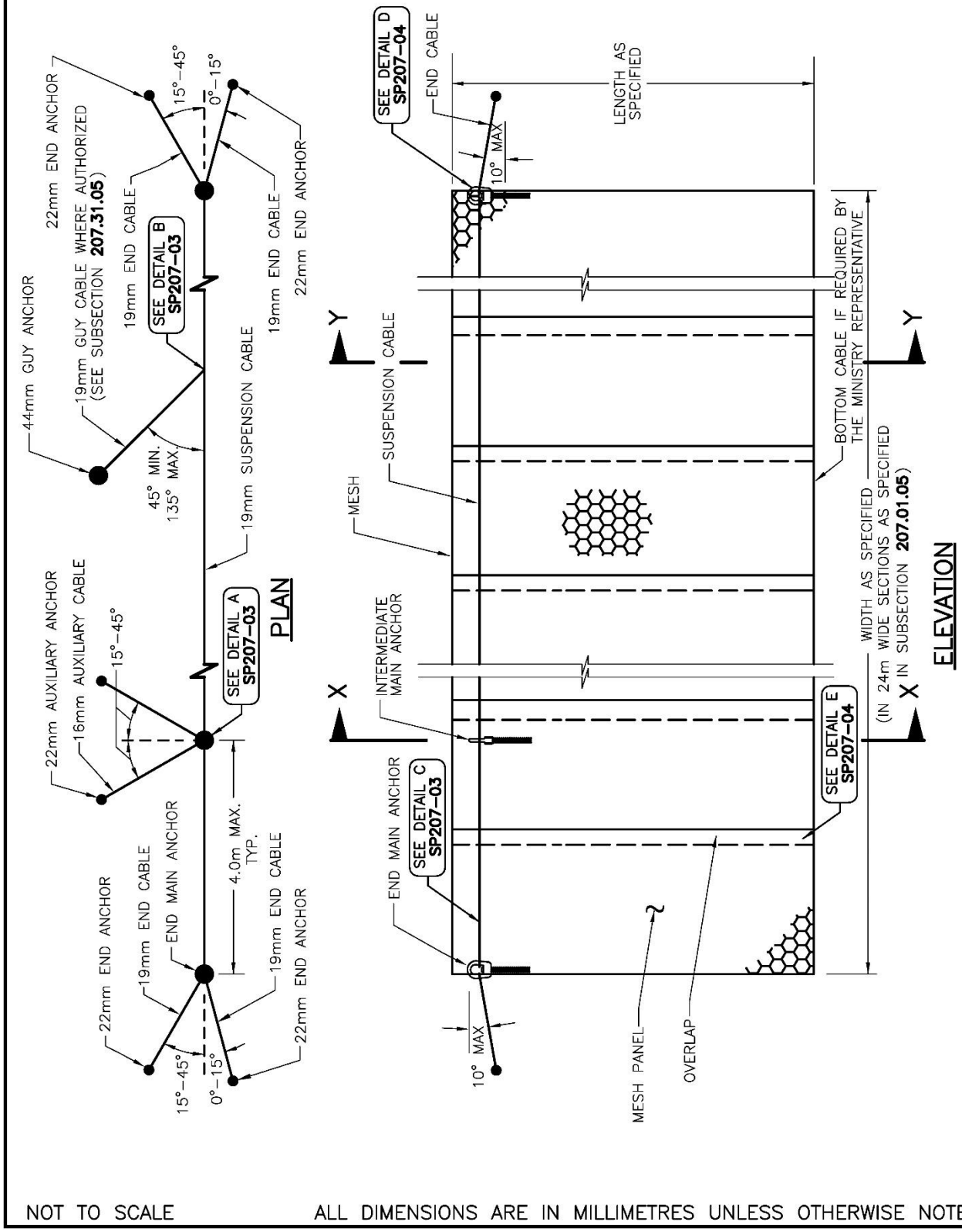
MEASUREMENT

207.81 Slope Mesh – Slope mesh shall be measured by the SQUARE METRE of slope meshed area.

PAYMENT

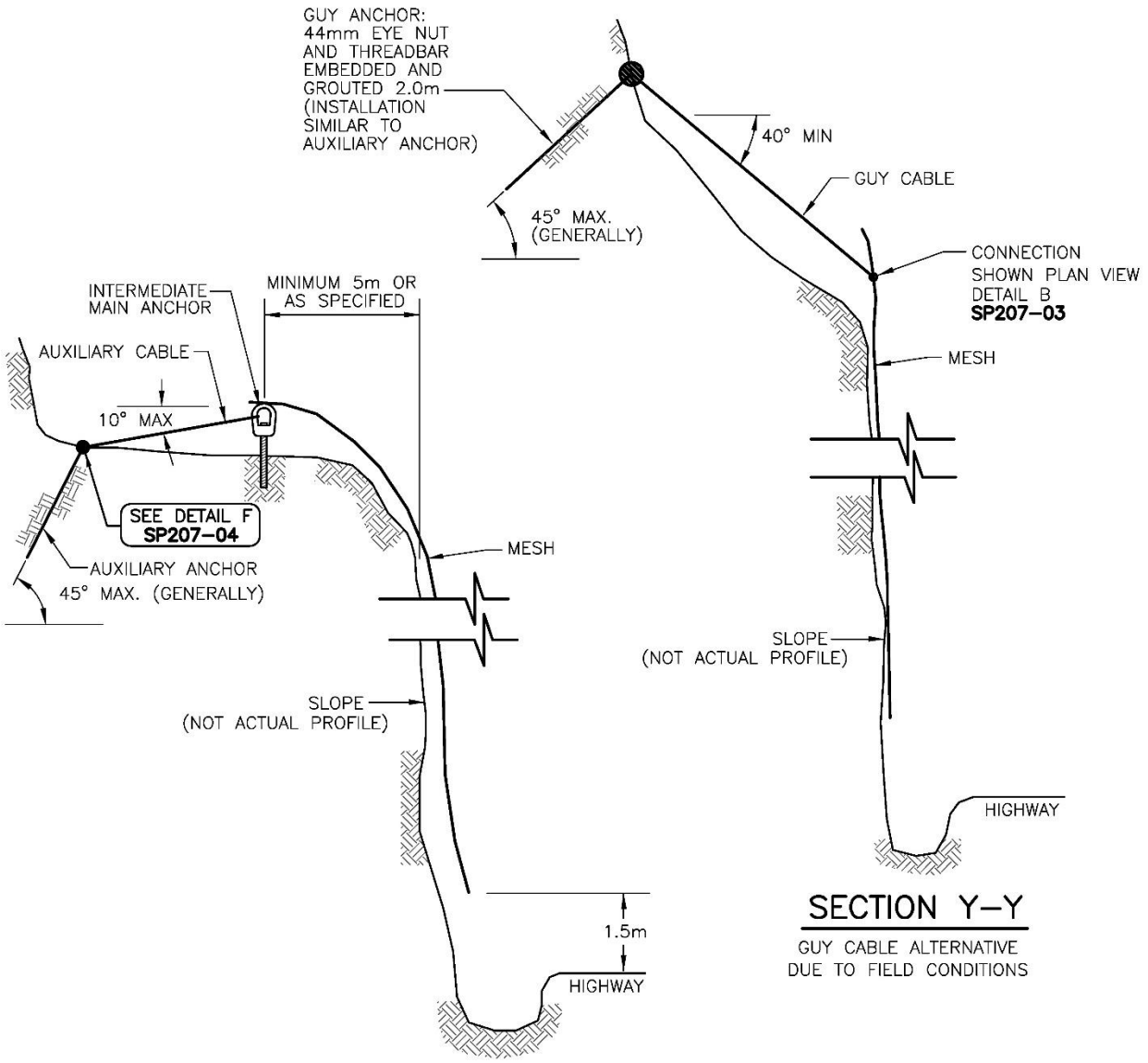
207.91 Slope Mesh – Payment for SLOPE MESH will be at the Contract Unit Price per square metre installed. The Unit Price will be full compensation for all requirements in this specification. Partial payment may be authorized where all components have been installed in a portion of the designated slope meshing area. No separate payment will be made for mesh overlap.

SLOPE MESH LAYOUT	SP207–01



NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SLOPE MESH – SECTIONS	SP207–02



SECTION X–X

SIDE VIEW AT INTERMEDIATE
MAIN ANCHOR

SECTION Y–Y

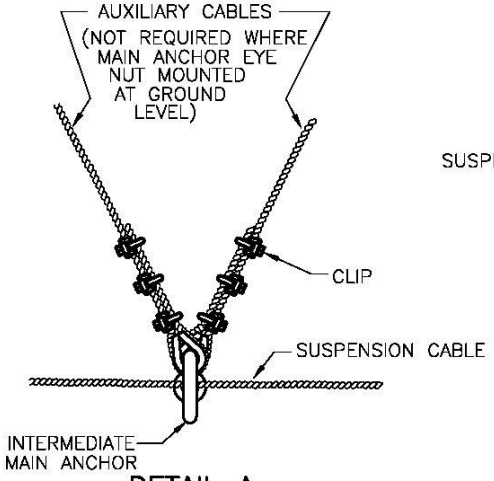
GUY CABLE ALTERNATIVE
DUE TO FIELD CONDITIONS

NOTES:
1. FOR LOCATION OF SECTIONS SEE DWG. **SP207–01.**

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

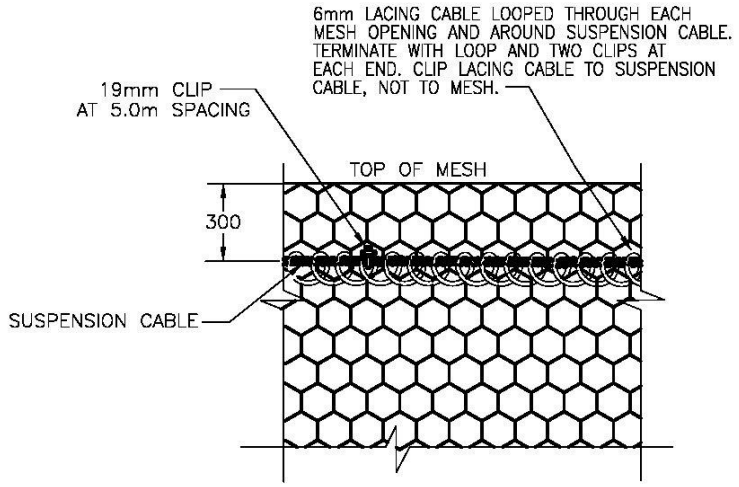
SLOPE MESH – DETAILS

SP207–03



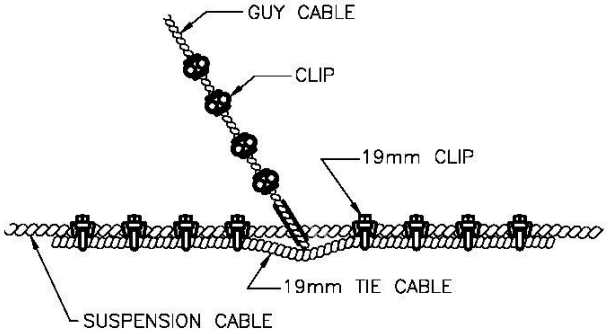
DETAIL A

INTERMEDIATE MAIN ANCHOR



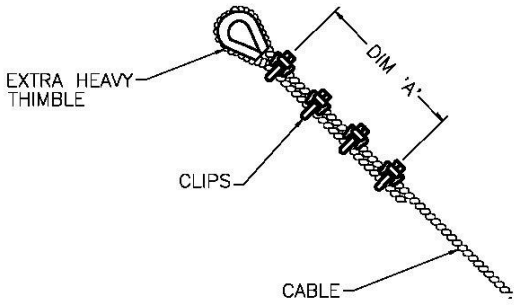
DETAIL C

MESH TO SUSPENSION CABLE CONNECTION



DETAIL B

GUY CABLE ALTERNATIVE DUE TO FIELD CONDITIONS



CABLE LOOP DETAIL

FOR SUSPENSION, END, GUY AND AUXILIARY CABLES (SEE TABLE 1)

TABLE 1: CLIPS

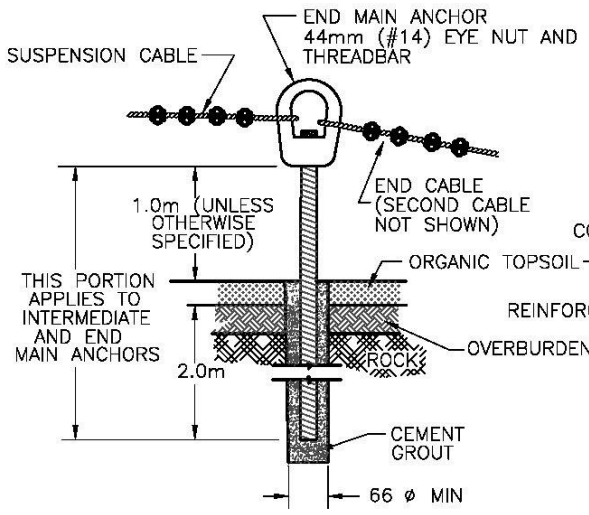
CABLE TYPE	CABLE SIZE (mm)	CLIP SIZE (mm)	N°. OF CLIPS FOR LOOP	DIMENSION 'A' (mm)	TORQUE Nm (ft. lbs)
Suspension, End, Guy	19	19	4	460	175 (130)
Auxiliary	16	16	3	300	113 (95)
Lacing	6	19	(varies)	N/A	54 (45)

NOTES:
1. FOR LOCATION OF DETAILS SEE DWG. SP207–01.

NOT TO SCALE
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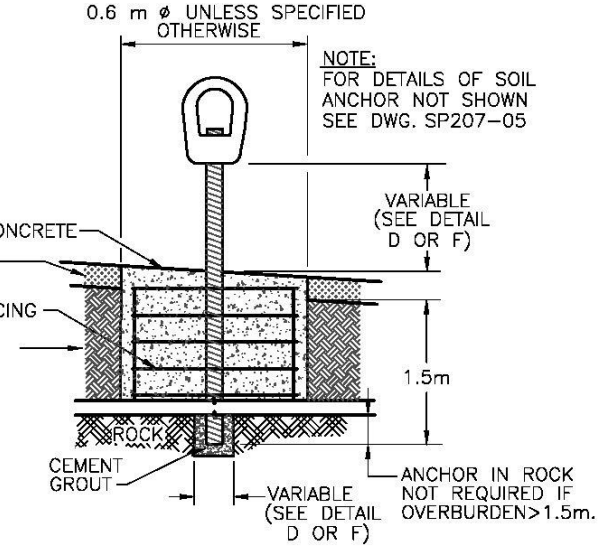
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SLOPE MESH – DETAILS	SP207–04



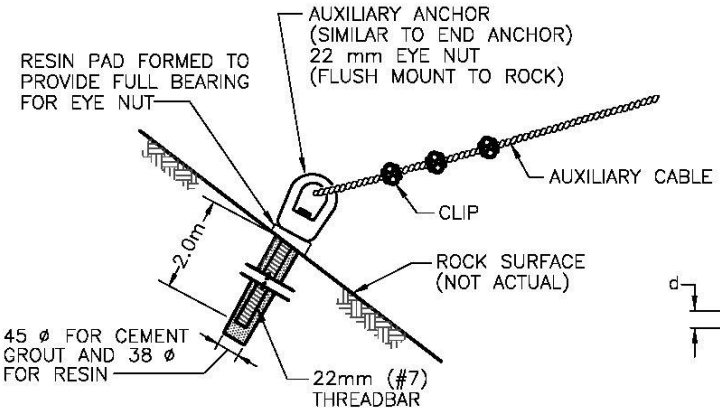
DETAIL D

END MAIN ANCHOR CONNECTIONS
FOR OVERBURDEN THICKNESS LESS THAN 0.6m
(MESH OMITTED FOR CLARITY)



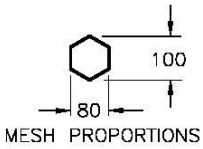
SOIL ANCHOR DETAIL

ALTERNATE ANCHOR DETAIL IN OVERBURDEN.
FOR OVERBURDEN THICKNESS GREATER
THAN 0.6m.

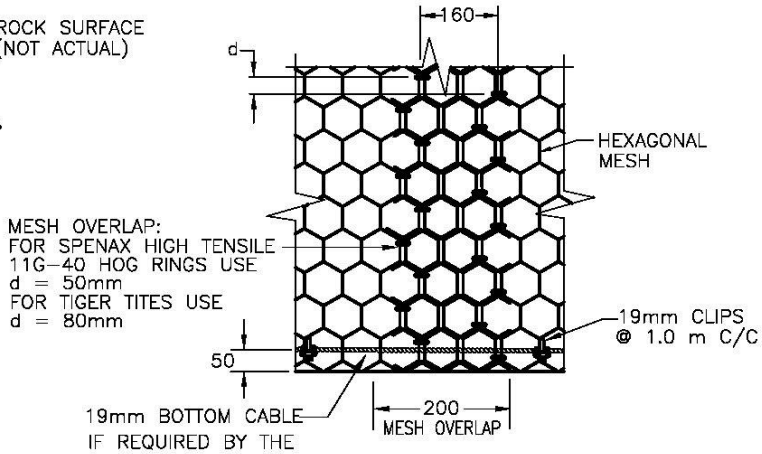


DETAIL F

AUXILIARY ANCHOR
CONNECTIONS



MESH PROPORTIONS



DETAIL E

MESH SEAM CONNECTIONS AND BOTTOM CABLE

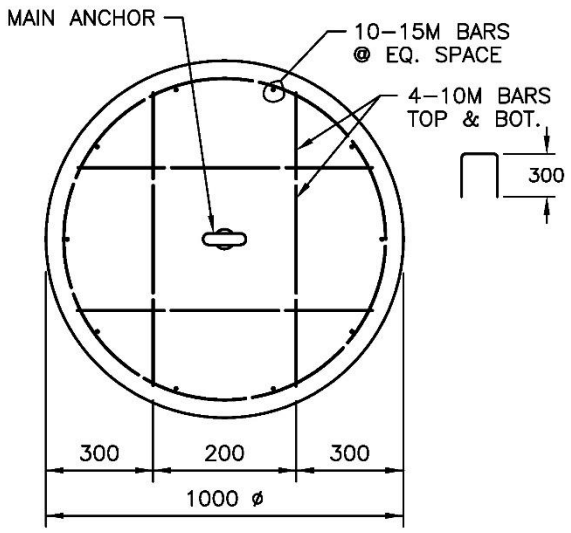
NOTES:
1. FOR LOCATION OF DETAILS SEE DWGS. SP207–01 & 02.

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

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SOIL ANCHOR CONCRETE AND REINFORCEMENT FOR SLOPE MESH INSTALLATIONS

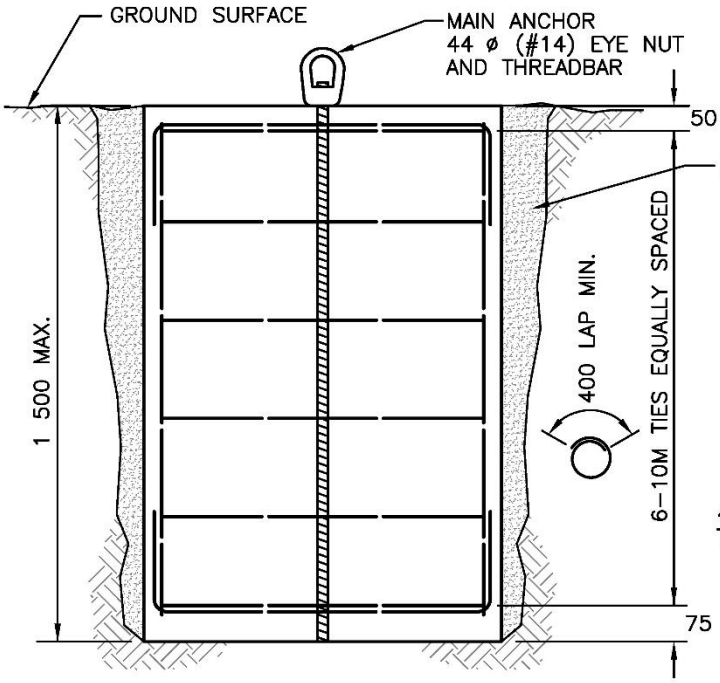
SP207–05



PLAN

NOTES:

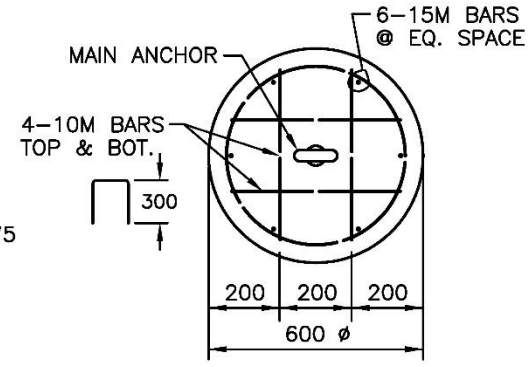
1. INSTALL 600 OR 1000mm ø SOIL ANCHORS AS REQUIRED.
2. REDUCE SOIL ANCHOR DEPTH WHERE ROCK IS ENCOUNTERED, AND INSTALL MAIN ANCHOR INTO ROCK TO PROVIDE TOTAL 1.5m EMBEDDED LENGTH.
3. CONCRETE REQUIREMENTS SHOWN ON SLOPE MESH DRAWING.
4. ALL REINFORCING STEEL TO MEET C.S.A. SPECIFICATION G30.18–M, GRADE 400R.
5. ALL REINFORCING STEEL TO HAVE 50mm COVER UNLESS SPECIFIED OTHERWISE.
6. PLACE REINFORCING BARS IN ACCORDANCE WITH RECOMMENDED PRACTISES OF THE CONCRETE REINFORCING STEEL INSTITUTE (C.R.S.I.)
7. CONCRETE AND REINFORCING STEEL TO MEET STANDARD SPECIFICATION SECTIONS **412** AND **218** UNLESS SPECIFIED OTHERWISE.
8. BACKFILL TO BE NATIVE MATERIAL COMPACTED USING HAND TOOLS TO SIMILAR DENSITY AS SURROUNDING GROUND, TAKING CARE TO FILL VOIDS TO THE EXTENT POSSIBLE.



ELEVATION

1000 ø SOIL ANCHOR

(MAY BE REQUIRED FOR MULTIPLE ANCHOR LOCATIONS WHERE AUTHORIZED BY THE MINISTRY REPRESENTATIVE.)



600 ø SOIL ANCHOR

(ALL NOT SHOWN SIMILAR TO 1000 ø SOIL ANCHOR)

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 208
DRAINS FOR ROCK CUTS

DESCRIPTION

208.01 Scope – This Section covers the installation of lined drains in the designated areas to the specified depth. The exact locations of drain holes will be dependent on field conditions and will be detailed by the Ministry Representative.

MATERIALS

208.11 PVC Pipe Materials – Drain hole lining shall consist of Schedule 40 slotted PVC pipe. The outside diameter of the pipe shall not be less than 67% of the borehole diameter, and the internal diameter of the pipe shall be a minimum of 19 mm. The slots shall be along the entire length of the pipe in one row. Unless otherwise specified, the slots shall be 0.5 mm wide, spaced 6 mm apart and have a minimum length of 30% of the outside circumference of the pipe.

CONSTRUCTION

208.31 Drilling – Advancement of drilling for drain holes shall be in an upslope direction. The dip angle for drain holes shall be five degrees (5°), or as specified by the Ministry Representative. Finished holes shall be flushed with air to remove deleterious materials.

208.32 PVC Pipe Installation – PVC pipes shall be installed into the finished holes immediately after flushing. Pipes shall be coupled according to the manufacturer's specifications. The pipe shall be installed along the entire length of the holes and shall protrude 0.30 m out of the rock face. The slots shall be on top when the pipe is installed. The pipe shall be secured in the hole, using plastic or wooden wedges driven into the hole collar if necessary, so that the pipe cannot be pulled out by hand.

MEASUREMENT

208.81 Lined Drains – Lined drains will be measured by the METRE of drain hole drilled.

PAYMENT

208.91 Lined Drains – Payment for LINED DRAINS will be at the Contract Unit Price per metre. Payment for lined drains will be authorized after installation to the Contract specifications. The Unit Price will be considered full compensation for all work and materials necessary to complete the installation prescribed in this Section.

SECTION 209
SHOTCRETE FOR ROCK CUTS

DESCRIPTION

209.01 General – Unless otherwise specified, the shotcrete standards of American Concrete Institute (ACI) 506.2 Specification for Shotcrete and the standards of good practice in ACI 506R Guide to Shotcrete shall apply. SS 211, Portland Cement Concrete, shall also apply for shotcrete, unless otherwise specified.

Shotcrete operations shall include, but not be limited to pre-construction and construction work including preparation of the mix design(s), quality control, surface preparation; installation of wick drains, weep holes and pipes; supply and installation of the shotcrete; preconstruction and construction testing, and curing; the supply, transportation and testing of shotcrete panels; any core extraction of the in-place shotcrete, and any other testing to determine curing or protection requirements.

209.02 Quality Control

209.02.01 General – The Contractor shall engage at its own expense, a specialist firm (the Contractor's Consultant) to provide on-site quality control for shotcrete operations. The Contractor's Consultant shall not be an employee of the Contractor.

209.02.02 Qualifications – The Contractor's Consultant shall have designed shotcrete for at least three (3) projects over the last five (5) years. The Contractor's Consultant's representative conducting on-site shotcrete tests shall be an ACI or Canadian Council of Independent Laboratories (CCIL) certified concrete technician.

209.02.03 Specific Requirements – For wet and dry process shotcrete, quality control shall include, but not be limited to the following:

(a) Observation of surface preparation before shotcrete application.

(b) Observation and testing during shotcrete application.

(c) Observation of curing operations.

209.02.04 Other Requirements – The Contractor's Consultant shall be present at other times as are appropriate to assist the Contractor in the preparation, assessment and adjustment of the various procedures to be used for shotcrete operations.

209.02.05 Field Report – The Contractor shall ensure that the Contractor's Consultant prepares a field report for each site visit, signed by the Contractor's Consultant. The report shall include details of the progress of shotcrete operations, any recommendations made to the Contractor and any problems encountered by the Contractor.

209.03 Submittals

209.03.01 The Contractor shall submit for review the following documentation by the pre-construction meeting:

(a) **Qualifications and Experience of the Contractor** – Provide a statement of the qualifications, experience and work function of all personnel assigned to shotcrete duties. A statement of previous work experience on similar projects shall also be provided. This statement shall include the project name, location, type of shotcrete, volume of shotcrete, year constructed and the owner contact name. The company, the nozzle operator and the predampener/pot operator shall each have a minimum of five (5) years demonstrated experience on shotcrete work.

(b) **Qualifications and Experience of the Contractor's Consultant** – The qualifications shall include the name of the consulting firm, and the name and qualifications of the Consultant's on-site representative who will be providing the quality control. The following information shall be included in the qualifications submitted:

(i) Project name, location and experience.

(ii) Contact name and phone number of the owner who can verify the experience of the Contractor's Consultant's site representative.

(c) **Proposed Laboratory Testing Agency** – Provide a statement of the testing agency's experience in performing laboratory tests on shotcrete. The agency shall be an independent, qualified laboratory with a proven record of performing tests on shotcrete on at least five (5) previous projects. The testing agency shall be certified by CCIL to CSA A283 *Qualification Code for Concrete Testing Laboratories*.

(d) **Certification of the Concrete Technician** – Provide evidence of ACI or CCIL Certification for the concrete technician who will be providing on-site shotcrete tests.

209.03.02 Documentation – The Contractor shall submit the following documentation at least ten (10) days prior to commencing shotcrete operations to the Ministry Representative:

(a) **Proposed Shotcrete Operations** – Provide a description of the proposed shotcrete operations including slope preparation method, shotcrete application method, and shotcrete curing and protection methods.

(b) Materials Records – Provide records showing source and proof of conformance to project specifications of the following materials:

- **(i)** Portland Cement (provide mill certificates)
- **(ii)** Silica Fume (provide mill certificates)
- **(iii)** Mix Water
- **(iv)** Aggregates
- **(v)** Admixtures (provide supplier data sheets)
- **(vi)** Fibre Reinforcement (if specified), include load/deflection curves to ASTM C1609 for the proposed mix design
- **(vii)** Mesh Reinforcement (if specified, provide mill certificates)

(c) Previous Performance Data – Provide previous performance data for the proposed shotcrete mix design, if available.

(d) Proposed Shotcrete Mix Design (Proportions) – Provide documentation for the proposed shotcrete mix design. Include such back-up data as requested. The proposed mix design submittal shall include:

- **(i)** Mix design number.
- **(ii)** Batch quantities in kg/m³ based on aggregates in an SSD (Saturated Surface Dry) moisture state for wet process shotcrete and ready mix supplied dry process shotcrete. Batch quantities for dry bagged supply shall be based on mass of aggregates in a dry state.
- **(iii)** A total cementitious content of not less than 20% by mass of the total ingredients.
- **(iv)** Aggregate source, bulk density, absorption, combined gradation data and a grain size distribution graph scaled between 0.075 mm and 20 mm showing the percent passing curve for the combined aggregate.

(e) Proposed Shotcrete Imprint – Provide a drawing of the proposed shotcrete imprint including content, imprinting method and materials that will be used to produce the imprint.

209.03.03 The Contractor shall submit for review the following documentation within 1 day following each day of shotcrete application:

(a) Field Report – Provide original field reports signed by the Contractor's Consultant.

(b) Daily General Records – Provide daily shotcrete records with the following information (certified by the Contractor's Consultant in the case of wet process shotcrete):

- **(i)** Contractor Name
- **(ii)** Nozzleman Name
- **(iii)** Helper Name
- **(iv)** Contractor's Consultant's Name (for wet process shotcrete)
- **(v)** Contractor's Consultant's Representatives Name (for wet process shotcrete)
- **(vi)** Date
- **(vii)** Weather
- **(viii)** Contractor's start and end time
- **(ix)** Shotcrete equipment type
- **(x)** Temperature
- **(xi)** Type of shotcrete (wet or dry process)
- **(xii)** When slump is measured (before or after admixture and fibres)

(c) Shotcrete Batch Records – Submit the following information for each batch of shotcrete:

- **(i)** Batch number
- **(ii)** Location of shotcrete
- **(iii)** Mix design number
- **(iv)** Batch time at plant (wet process)
- **(v)** Batch arrival time on-site
- **(vi)** Truck departure time (wet process)
- **(vii)** Placement start time
- **(viii)** Placement end time
- **(ix)** Quantity used
- **(x)** Water/cementitious materials ratio (wet process)
- **(xi)** As-shot air content (wet process)
- **(xii)** Slump (wet process) – indicate when measurement is made (e.g.: after air entrainment is added, before superplasticizer is added and before fibre reinforcement is added)
- **(xiii)** Percent rebound (visual estimate)
- **(xiv)** Average thickness of shotcrete
- **(xv)** Test panel number
- **(xvi)** Shotcrete delivery tickets from batch plant (wet process)
- **(xvii)** Comments on the quality of the shotcrete and any problems that arise
- **(xviii)** Proposed revised mix design

209.03.04 The Contractor shall submit the laboratory test results within two (2) working days after testing to the Ministry Representative. These test results shall compare

the shotcrete performance with the requirements given in Table 209-A.

Records shall include all reporting requirements outlined for construction testing. Records shall also include an original summary report of all test results, showing the sample number, sample origin, panel number, panel origin, date the panel was sprayed, panel shipping date, date panel received at the laboratory, photos of prepared test samples, test date, sample age at testing, test results for each sample, average test results of each set of three samples taken pursuant to SS 209.39 for each test at a given age, and acceptability of the results.

209.04 Definitions – The following definitions refer to words and terms used in this specification. For definitions not covered in this document, refer to ACI 506R and ACI 506.2.

(a) **Blow pipe** – Air / water jet operated by nozzle operator's helper during shotcrete placement to assist in keeping rebound and overspray out of the work.

(b) **Dry Process Shotcrete** – A process where most of the water added to the shotcrete mixture is added at the nozzle.

(c) **Overspray** – Shotcrete material deposited away from the intended receiving surface.

(d) **Pot (Sometimes called Gun)** – Machine used to meter the shotcrete into the hose.

(e) **Predampener** – Device used to mix a portion of the mixing water with the shotcrete. Can also be used to mix accelerator with shotcrete.

(f) **Saturated Surface Dry** – The state of a material in which it will neither add nor subtract moisture from other material placed in contact with it.

(g) **Slake** – A process of deterioration of freshly exposed rock due to exposure to the atmosphere and/or water.

(h) **Sloughing (also called sagging)** – Subsidence of shotcrete, due generally to excessive water in the mix or placing too great a thickness of shotcrete in a single pass.

(i) **Weep Pipes** – Plastic pipes inserted in water bearing fissures to relieve water pressure from behind the shotcrete.

(j) **Wet Process Shotcrete** – Shotcrete in which all of the ingredients are mixed before introduction into the delivery hose. Compressed air is introduced to the material flow at the nozzle. If an accelerator is used, it is normally added at the nozzle.

MATERIALS

209.11 Materials

209.11.01 Cement – Cement shall be Type GU or HE, or where approved by the Ministry Representative, Type GUL. GUL cement may be used only if:

(a) Shotcrete is not subject to very severe sulphate exposure conditions on Site (CSA A23.1 exposure condition S-1); and

(b) GUL cement, silica fume, and any other supplementary cementitious materials and additives proposed to be use are assessed for compatibility, and collectively will not produce any deleterious effects on the long-term properties of the shotcrete or its bond to the substrate.

209.11.02 Silica Fume – Silica fume shall meet the requirements of CSA-A23.5 [See CSA A3000], Supplementary Cementing Materials, Type U Silica Fume, with the following additional requirements:

- Minimum SiO_2 content – 90 percent (90%) by mass
- Maximum carbon content: 5 percent (5%) by mass

The Contractor may propose alternatives to silica fume, which shall be subject to approval by the Ministry Representative.

209.11.03 Pozzolan – Pozzolan shall be Type F flyash.

Table 209-A: Shotcrete Performance Requirements

Test Description	Test Method	Age (Days)	Specified Requirement
*Maximum Water / Cementitious Materials Ratio		-	0.45
Air Content – As Shot, %	CSA A23.2-4C	-	4 ± 1½
*Slump at discharge into shotcrete pump, mm	CSA A23.2-5C	-	80 ± 30
Minimum Compressive Strength, MPa	CSA A23.2-14C	7 28	30 40
Minimum Flexural Strength, MPa	ASTM C1609	7	4
Minimum Toughness Performance Level (TPL)	ASTM C1609 and this specification	7	III
Maximum Boiled Absorption, %	CSA A23.2 – 11C	7	8
Maximum Volume of Permeable Voids, %	ASTM C642	7	17

*Notes: *Properties apply to wet process shotcrete only.*

209.11.04 Water – Surface or ground water may be used for cleaning the slope and curing provided it is free of deleterious substances. Verification that the water intended for use is acceptable is at the discretion of the Ministry Representative and at the expense of the Contractor.

209.11.05 Aggregates – Aggregate shall be normal density. The combined aggregate gradation limits shall meet the limits shown in Table 209-B.

Table 209-B: Combined Gradation Limits for Shotcrete Aggregate

U.S. STD Square Mesh	Metric Sieve Size (mm)	Total Passing Each Sieve Size (% By Mass)
1/2	14	100
3/8	10	90 – 100
#4	5	70 – 85
#8	2.5	50 – 70
#16	1.25	35 – 55
#30	0.630	20 – 35
#50	0.315	8 – 20
#100	0.160	2 – 10

209.11.06 Admixtures

(a) General – Chemical admixtures shall include air-entraining, water reducing and set retarding admixtures where appropriate.

(b) Superplasticizers – Superplasticizers shall meet the requirements of ASTM C1141.

(c) Accelerating Admixtures in General – Accelerating admixtures shall only be used with the written consent of the Ministry Representative.

(d) Accelerating Admixture Requirements – Accelerating admixtures:

(i) shall be non-chloride based accelerating admixtures meeting the requirements of CSA A23.1M Clause 15.1.7.1;

(ii) shall be compatible with the cement being used when tested in accordance with ASTM C1141;

(iii) shall have a demonstrated ability, when used at the required proportions, to produce accelerated shotcrete meeting the performance criteria specified in Table 209-A;

(iv) wet process shotcrete shall be liquid, dispensed at the nozzle in closely controlled quantities;

(v) dry process shotcrete shall be liquid, dispensed at the nozzle in closely controlled quantities; or be powdered and either:

(A) Integrally mixed in the case of dry bagged material or;

(B) Mechanically proportioned through the predampener in the case of ready mix supply; metering of accelerators directly into the pot will not be permitted.

209.11.07 Fibre Reinforcement – Unless otherwise specified, shotcrete reinforcement shall consist of steel or polyolefin fibres. The fibres shall be capable of meeting the performance criteria specified in Table 209-A and the following requirements:

(a) Steel Fibres – Fibres shall meet the requirements of ASTM C1116 Type I.

(b) Polyolefin Fibres – Fibres shall only be used in wet process shotcrete and shall meet the requirements of ASTM C1116 Type III.

209.11.08 Mesh Reinforcement – Where specified instead of or in addition to fibre reinforcement, shotcrete shall be reinforced with mesh anchored to the surface before shotcrete application. For fibre reinforced shotcrete, additional reinforcement may be required by placing mesh panels up to 0.60 m by 0.60 m without anchors between successive shotcrete lifts where final shotcrete thicknesses are expected to exceed 300 mm in local areas.

(a) Mesh Materials – Mesh shall consist of welded wire fabric 102 mm by 102 mm opening (4 by 4 inch) by MW13.3 / MW13.3 (8 gauge) wire meeting ASTM A1064 and hot dip galvanized meeting ASTM A123.

(b) Mesh Anchors – Anchors shall consist of Grade 517 MPa steel, conforming to CSA G30.18, minimum 19 mm nominal diameter.

Face plates shall be 150 mm by 150 mm by 6 mm. Face plates, nuts and washers shall be capable of developing full anchor pullout capacity.

(c) Mesh Anchorage – Mesh anchors shall be secured by resin or grout.

(i) Grout – Grout shall have the following properties:

(A) be pre-mixed, unsanded, non-metallic, and non-shrink cementitious grout containing silica fume.

(B) may be mixed to a flowable consistency, typically with a water to cement ratio of 0.35 or as specified by the grout manufacturer to achieve or exceed the required strength.

(C) minimum 7-day compressive strength of 30 MPa and a minimum 28 day compressive strength of 40 MPa, tested in accordance with CSA A23.2-09.

(D) admixtures to be used according to the manufacturer's specifications and as authorized by the Ministry Representative.

(E) calcium chloride-based accelerators are not permitted.

(ii) Resin – Resin shall have the following properties:

(A) be supplied in cartridge form.

(B) a minimum unconfined compressive strength of 90 MPa when fully mixed and cured and tested in accordance with CSA A23.2-09.

(C) be encased in a plastic film that provides optimum resistance to moisture and is easily ruptured to enable complete mixing during installation.

(D) suitable thixotropic and viscous properties to permit adequate mixing of the resin components by rotation of the rock bolt bar and to contain the resin within the drill hole.

(E) an easily identifiable gel time recommended by the resin manufacturer.

(F) reach 80% of its ultimate strength within a time interval equal to five times the gel time.

(G) be non-shrink after the gel time.

(H) be unaffected by mild acids or mild alkalis.

(I) be supplied in cartridge boxes labelled with the resin expiry date.

(d) Mesh Coverage and Overlap – Mesh shall be installed over the entire area to be shotcreted, unless otherwise specified. Mesh joints shall be lapped 1.5 mesh openings in each direction (horizontally and vertically) and tied at minimum 100 mm intervals with 16 gauge tie wire to achieve full contact between mesh layers.

(e) Mesh Anchor Spacing – Mesh shall be anchored on minimum 1.2 m centres in each direction or closer where required to form the mesh to within 50 mm of the slope surface. The mesh shall be in full contact with the surface where practicable. The mesh may also be anchored to any previously installed rock bolts provided the spacing is consistent with the spacing required for the above anchors.

(f) Anchor Installation – Anchors shall be installed as recommended by the anchor manufacturer. The minimum anchor embedment length shall be 1.0 m or as specified. Resin or grout shall be applied to full anchor embedment.

(g) Mesh Anchor Testing – A total of 5% of the anchors, selected by the Ministry Representative, shall be load tested to 186 kN. This load shall be held for 10 minutes. A load loss of greater than 10% of the applied load shall be indicative of anchorage failure. Any failed anchors shall be replaced at the Contractor's expense.

209.11.09 Anchor Reinforcement – Where specified, anchor reinforcement shall be installed. Anchor materials, installation and testing shall meet the requirements for mesh anchors unless otherwise specified. Face plates for these anchors shall be "Spider Plates" conforming to SS Drawing SP209-02. These anchors and plates shall be fully embedded in the shotcrete.

209.11.10 Materials Handling and Storage

(a) General – All materials shall be stored and handled in a manner that will prevent damage, deterioration or contamination.

(b) Hot Weather Storage – During hot weather prebagged material shall be stored in a manner that will allow maximum cooling of the material.

(c) Cold Weather Storage – During cold weather prebagged material shall be stored in a manner that will provide maximum heat retention of the material.

209.12 Performance Requirements

209.12.01 General – Shotcrete shall meet the performance requirements in Table 209-A.

209.12.02 Air Content – The air content shall be determined on wet process shotcrete sprayed into a CSA-A23.2-4C air pressure meter base or freshly applied material removed from in-place and consolidated into the air meter base by rodding.

CONSTRUCTION

209.31 Batching, Mixing and Conveyance

209.31.01 General – Shotcrete shall be batched, mixed and supplied using (one of) the following system(s):

(a) dry bagged premix supply to ASTM C928 or:

(b) ready mix supply

209.31.02 Mixing – Shotcrete supplied shall be uniformly mixed with no evidence of segregation or improper mixing.

209.31.03 Batching and Conveyance – Ready mix supplied shotcrete shall be batched and conveyed at a rate that allows a consistent application of shotcrete. Irregularities of conveyance that result in delays or a surplus of aged material on-site shall be avoided. Shotcrete that is

more than 90 minutes old from the time of batching will not be accepted, unless hydration controlling admixtures are used to maintain suitable slump. Shotcrete that has stiffened excessively shall not be used.

209.32 Pre-construction Testing

209.32.01 General – The Contractor shall perform pre-construction shotcrete tests (prior to full-scale shotcrete operations), at a shotcrete test site chosen by the Ministry Representative, to demonstrate the competence of the crew, the adequacy of the equipment and the adequacy of the shotcrete mix design. The requirement for pre-construction testing may be waived by the Ministry Representative if sufficient documentation of previous satisfactory performance on similar work is provided.

209.32.02 Previous Performance – Documentation for previous satisfactory performance shall meet the following requirements:

(a) **Company** – The company shall have applied more than 100 m³ of shotcrete on a minimum of 3 projects in the last 5 years.

(b) **Nozzle Operator** – The nozzle operator shall have applied more than 100 m³ of shotcrete on a minimum of 3 projects in the last 5 years.

(c) **Predampener/Pot Operator** – The predampener/pot operator shall have mixed more than 100 m³ of shotcrete on a minimum of 3 projects in the last 5 years.

(d) **Contacts** – The contact name and number for each project above shall be supplied upon request.

209.32.03 Application – Shotcrete applied at the shotcrete test site shall be applied in accordance with the general shotcrete provisions in these specifications.

209.32.04 Test Panel – One test panel shall be shot and tested for the pre-construction shotcrete testing as described for construction testing.

209.32.05 Test Results – The Contractor shall not apply shotcrete outside of the shotcrete test site area until the pre-construction shotcrete test results have been evaluated by the Ministry Representative and authorization is given to proceed with the work.

209.32.06 Revised Methods – If the results of the shotcrete tests are unsatisfactory in the opinion of the Ministry Representative, then, notwithstanding the Ministry Representative's prior review of the Contractor's pre-construction submittals, the Contractor shall adopt such revised methods as are necessary to achieve the required results.

209.32.07 Additional Tests – The Contractor shall perform, at any time during the progress of the work, additional shotcrete tests at designated test sites if the shotcrete applied in the areas specified in the Contract are unsatisfactory in the opinion of the Ministry Representative.

209.33 Weather Conditions

209.33.01 Heavy Rain or High Wind – During periods of heavy rain or high wind, the Contractor shall ensure that the quality of the finished shotcrete product is not compromised. Shotcrete shall not be applied during such weather unless protective measures can be practicably taken.

209.33.02 Hot Weather – During periods of hot weather, when ambient temperatures are expected to rise over 30°C, the following requirements shall be met

(a) **Curing** – The Contractor shall implement curing immediately after the shotcrete has been placed.

(b) **Temperature at Placement** – The Contractor shall take any steps necessary to ensure that the temperature of the shotcrete supplied to the site is less than 25°C at the time of placement.

209.33.03 Cold Weather – During periods of cold weather, where the minimum ambient air temperature falls below 5°C, the following requirements shall be met:

(a) **Temperature at Placement** – The Contractor shall take whatever steps are necessary to ensure that the in-place shotcrete temperature is between 10°C and 25°C at the time of placement.

(b) **Initial Curing Temperature** – The Contractor shall ensure that, during the initial curing period (4 days), the in-place shotcrete temperature will not fall below 5°C. Shotcrete shall not be placed unless measures can be practicably taken to ensure these requirements are met.

(c) **Temperature at the Pot** – The Contractor shall not allow the temperature of the dampened dry process shotcrete at the pot to exceed 30°C.

(d) **Ice, Snow or Frozen Surfaces** – Shotcrete shall not be applied to frozen surfaces, or surfaces covered with frost, ice or snow.

209.34 Surface Preparation – Surfaces to which shotcrete is applied shall be cleaned of loose rock, dirt and debris which if left in place could impair the bond of shotcrete to rock, concrete or existing shotcrete. Water or air and water jets shall be used to clean rock surfaces. Compressed air only shall be used to clean rock surfaces that have a tendency to slake from contact with water.

209.35 Drainage – Unless otherwise controlled, water seepage from slopes shall be controlled by installing weep holes, weep pipes, and wick drains. Surface runoff shall be controlled using suitable water diversion systems.

209.35.01 Weep Holes – Weep holes (and weep pipes as required) shall be installed before shotcrete is applied. These weep holes shall be on 1 m to 1.5 m centres in the

areas to be shotcreted in joints, fractures and faults, or at a closer spacing where required by the Ministry Representative. Holes and pipes shall be inclined to permit free drainage.

209.35.02 Weep Pipes – To prevent blockage of weep holes due to entry of shotcrete or rebound materials, weep pipes shall be secured in weep holes and temporarily plugged at the outer end of the pipe. The plugs shall be removed when shotcreting is completed. Weep pipes shall consist of 25 mm minimum outside diameter Schedule 40 plastic pipe.

209.35.03 Active Seepage Areas – Where active seepage occurs, weep holes shall be drilled approximately 150 mm or more to intercept fractures or openings wherever practicable and weep pipes shall be installed prior to the application of shotcrete. Where drilling is not practicable, the weep pipes shall be secured in the fractures by other appropriate means.

209.35.04 No Active Seepage Areas – Where no active seepage occurs, weep holes shall be installed in joints, fractures and faults by one of the following methods:

(a) by securing the weep pipes by appropriate means before shotcreting;

(b) by marking locations using sticks or other appropriate means before or during shotcreting and drilling at the marked locations after shotcreting. Weep holes shall be 25 mm diameter and shall be drilled to intersect the joints, fractures and faults under the shotcrete at the marked locations. Weep pipes are not required for such holes.

209.35.05 Weep Pipe Removal or Cutting – Weep pipes shall be removed or cut level with the shotcrete surface after the shotcrete has reached initial set, within one day after application.

209.35.06 Additional Weep Holes – Additional weep holes shall be drilled after shotcreting where there are visible signs of water pressure, such as seepage and damp spots in the hardened shotcrete.

209.35.07 Wick Drains – In areas of active seepage, wick drains shall be installed before shotcreting in addition to weep holes if requested by the Ministry Representative. Wick drains shall consist of a flexible polypropylene core drain wrapped with a proven durable geotextile filter fabric. Wick drains shall be 3 mm by 100 mm size (variable length), with #120 sieve filter, 0.17 mm/s filter permeability, and 0.8 kN core grab tensile strength. They shall be able to withstand all handling, abrasion and distortion that occur during installation. Wick drains shall be secured to the surface on maximum 0.5 m centres with pneumatically driven nails, or authorized substitute fasteners. Fastener spacing shall be reduced where required to ensure wick drains are in full contact with the surface. Wick drains shall be installed and covered with shotcrete in a manner that allows gravity drainage to the toe of the slope.

209.35.08 Surface Runoff – Where the bond of shotcrete to rock may be impaired by surface water runoff, this water shall be controlled and directed away from the area to be shotcreted.

209.36 Existing Rock Bolts – Within or near areas designated for shotcrete where rock bolts were installed (not for the sole purpose of securing mesh) prior to shotcrete application, the rock bolts shall be protected or extended through the shotcrete as described in the following subsections.

209.36.01 Extension of Rock Bolts – Unless otherwise specified, where rock is fractured within 0.5 m of the existing rock bolts, the face plates and nuts shall be removed before shotcrete is applied. The bar shall be extended, where necessary, using a short piece of bar and a coupler. The shotcrete shall be applied up to and against the existing bar. The face plates and nuts shall then be reinstalled over the shotcrete after the shotcrete has cured. Each face plate shall be replaced to the same bar from which it was removed. The exposed bar length shall be 100 mm following reinstallation of the face plate. The bar shall be nominally tensioned (using a wrench to the nut).

209.36.02 Protection of Rock Bolts – All other rock bolts shall be protected from overspray. Where overspray was not anticipated, the affected rock bolts shall be cleaned off by appropriate means.

209.37 Application

209.37.01 General – Prior to application of any shotcrete, the Contractor shall obtain authorization to proceed after the areas prepared for shotcrete have been reviewed by the Ministry Representative. Review by the Ministry Representative shall not relieve the Contractor from the responsibility for ensuring that the shotcrete operations are conducted in a satisfactory manner in accordance with these specifications.

209.37.02 Wetted Surface – Not more than one hour prior to application of shotcrete, all surfaces (except slaking ground) to be shotcreted shall be sprayed with water. Wetted surfaces shall be allowed to dry back to a saturated-surface-dry condition prior to application of shotcrete. If necessary, a blow-pipe shall be used to facilitate removal of surface water. Compressed air used in the blow pipe shall be oil-free.

209.37.03 Shotcrete Layers – Wherever possible, shotcrete shall be applied to the full thickness in a single layer, provided it does not slough or become delaminated. If multiple lifts are required, the previous shotcrete lifts shall be scraped or broomed prior to the time of set to remove any loose material, rebound, overspray, laitance or any other material that may impair bond of subsequent layers. Previous lifts shall be prevented from drying out by moist curing. Construction joints shall be to a 45° edge.

209.37.04 Rebound and Overspray – A blow pipe shall be used during the course of the work to remove accumulations of rebound and overspray from areas to be shotcreted. Rebound and overspray shall not be incorporated in the completed work.

209.37.05 Thickness and Area Covered – In general, an average thickness of 100 mm of shotcrete shall be applied over fractures, joints and faults in the areas designated for shotcrete, unless otherwise specified by the Ministry Representative. In general, shotcrete shall cover laterally to a maximum of 0.50 m beyond fractures, joints and faults over solid rock or concrete areas.

209.37.06 Access – Suitable devices shall be employed to allow access to the work for shotcreting. Access shall be such that fresh shotcrete is not damaged by hoses, equipment or personnel.

209.37.07 Surface Finish – The shotcrete surface shall be left in a natural gun state unless otherwise specified.

209.37.08 Embedments – A cover of 50 mm of shotcrete shall be applied over any reinforcing steel or other embedments. If greater total thicknesses of shotcrete are applied to encapsulate the embedments this work shall be performed at no cost to the Ministry.

209.37.09 Imprint – The plastic shotcrete surface shall be neatly imprinted with capitalized lettering 75 mm high with the following information arranged in the order listed below:

- MoTT
- The shotcrete completion date, in the format MMM, YYYY
- The name of the Contractor

The imprint shall be located in a small area in the bottom right corner of each continuous shotcrete area, unless otherwise authorized by the Ministry Representative. Where practicable, the information shall be readable unaided from highway level following shotcrete curing. Imprints produced by scraping letters into the plastic shotcrete will not be acceptable.

209.38 Curing and Protection

209.38.01 Curing Compounds – Membrane curing compounds shall meet ASTM C309 and shall be applied if necessary and generally in accordance with the manufacturer's recommendations. The application rate of a curing compound on the rough shotcrete surface shall generally be twice the rate used for smooth concrete surfaces. Curing compounds shall not be used on the substrate or in areas where additional shotcrete will be applied. If a membrane curing compound is not used, then moist curing shall be implemented.

209.38.02 Moist Curing – The surface of the shotcrete shall be maintained in a moist condition for a minimum period of 4 days following shotcrete application or until an in-place compressive strength of 20 MPa has been reached. Wet burlap and polyethylene sheet or water sprays shall be used to provide moist curing.

209.38.03 Protection from Freezing – The shotcrete shall be protected from freezing for a minimum period of 7 days after placing or until an in-place compressive strength of 20 MPa has been reached, whichever occurs first.

209.38.04 Sampling and Testing – If compressive strength samples are used to determine curing or protection requirements, the cost for obtaining the samples, shipping the samples and performing the tests shall be the responsibility of the Contractor. Extraction of cores shall be in accordance with CSA A23.2-14C. Compressive strength testing shall be performed according to requirements for construction testing.

209.39 Construction Testing

209.39.01 On-Site Tests – For wet process shotcrete, the following on-site tests shall be conducted for each truck:

(a) As-placed air content; and

(b) Slump at discharge into pump.

209.39.02 Thickness Monitoring – The Contractor shall monitor the thickness of shotcrete applied using depth gauges or probes. Thickness shall be measured on 1.0 m centres or closer as required to provide thickness control.

209.39.03 Test Panels in General – The Contractor shall produce test panels in accordance with the requirements of ASTM C1140, but panels shall have minimum base dimensions of 500 mm x 500 mm and a minimum depth of 125 mm as shown in SS Drawing SP209-01. The form base shall be made from 19 mm thick sealed plywood, and side pieces shall be made from two by six timbers. The test panel molds shall have 45° sloped sides to permit escape of rebound.

209.39.04 Test Panel Frequency – One construction test panel shall be shot on each day of shotcrete placement or for every 50 m^3 of shotcrete being placed, whichever results in more panels. The time of shooting the panels may be chosen by the Ministry Representative.

209.39.05 Test Panel Shooting and Marking – Test panels shall be shot using the same nozzle and length of hose as the production shotcrete. The Ministry may request test panels to be shot at the location of the in-situ shotcrete. Construction test panels shall contain no reinforcement or embedments (other than fibre reinforcement). For each test panel the Contractor shall provide a record of the test panel number, date cast, date shipped and date received by the laboratory.

209.39.06 Test Panel Curing and Transportation – Test panels shall be field cured, in the forms, in the same manner as the specified shotcrete work, for a minimum period of 48 hours, prior to transport to the test laboratory. Panels

shall be covered with wet burlap and plastic sheet to prevent drying. In cold weather, panels shall be protected from temperatures below 5°C by covering with thermal insulation blankets or by placing panels in a suitably heated enclosure. Test panels shall be transported in their forms and kept moist, with care being taken not to crack or damage the shotcrete.

209.39.07 Test Panels in Laboratory – The test panels inside the forms shall be placed in a moist room in the laboratory, maintained at temperature of 23 ± 2°C and relative humidity of 98 ± 2%. After age 3 days the test panels shall be removed from the forms and returned to the moist room until the time of testing.

209.39.08 Test Specimen Extraction and Storing – At specified test ages, test specimens shall be extracted from the panels using either diamond sawing or coring. Test specimens shall be maintained in a moist state in the moist room in the laboratory or in a lime-saturated water bath maintained at 23 ± 2°C until the time of testing.

209.39.09 Compressive Strength Test Specimens – Compressive strength test specimens shall preferably be 75 mm diameter cores. Length/diameter ratios shall not exceed 2:1 and shall not be less than 1:1.

209.39.10 Compressive Strength Testing – For each panel, three specimens shall be tested for compressive strength at each specified age. Compressive strength tests shall be conducted in accordance with CSA A23.2-14C. Measured compressive strengths shall be corrected to equivalent 2:1 length:diameter cores, using the core correction factors given in CSA A23.2-14C (Table 1). Reporting of results shall be according to Section 5 (Reporting) in CSA A23.2-14C.

209.39.11 Flexural Strength and Toughness Test Specimens – Flexural strength test specimens shall have dimensions of 100 mm x 100 mm x 350 mm. Reporting of results shall be according to "11. Report" in ASTM C1609 and shall include all load – deflection graphs.

209.39.12 Flexural Strength and Toughness Testing – For each panel, three specimens shall be tested for flexural strength and toughness at the specified age. Flexural strength and toughness tests shall be conducted in accordance with ASTM C1609 with the following additional requirements:

(a) Measuring System – A "Japanese Yoke" or other measuring system that precludes any extraneous deflection measurements shall be used. An automatic data acquisition system or equivalent shall also be used during the tests.

(b) Accuracy of Measurement – The accuracy of the first crack deflection shall be verified by comparing the measured value with the theoretically calculated value by using the formula:

$$\delta = \frac{23PL^3}{1296EI} \times \left[1 + \frac{216H^2(1+\mu)}{115L^2} \right]$$

Where:

δ = mid-span deflection in mm

P = load at first crack in kN

L = load span in mm

E = modulus of elasticity (for convenience usually based on an estimated compression modulus of elasticity) in GPa

I = moment of inertia in mm^4

H = height of specimen in mm

μ = Poisson's Ratio (for convenience usually based on an assumed compression Poisson's Ratio of 0.2)

For a 100 mm x100 mm x 350 mm beam tested on a 300 mm span the first crack deflection for typical fibre reinforced shotcrete will have first crack deflection in the range of about 0.03 mm to 0.05 mm. Extraneous deflections have been eliminated if first crack deflections are within this range.

(c) Recording of Deflections – An LVDT connected to an x-y plotter or digital recording shall be used to capture load vs. deflection response. Dial gauges shall not be used for recording deflections. Bending Load shall be plotted against Net Midspan Deflection.

(d) Testing Machine – A deflection-control testing machine shall be used, but shall not be of the closed-loop or soft testing type.

(e) Total Deflection – Deflection measurements shall continue until a total deflection of 2 mm is reached.

209.39.13 Toughness Performance Level Data Analysis – Toughness Performance Level of flexural strength specimens shall be obtained by the following analysis:

(a) Template – A template shall be created using the percent of design flexural strength shown in Table 209-C, but expressed in load equivalent units.

The following formula shall be used to determine load equivalent units;

$$P_d = \frac{\sigma \times B \times H^2}{L} \times \frac{1}{1000}$$

where:

P_d = design flexural load in kN

σ = design flexural strength in MPa

B = beam width in mm

H = beam depth in mm

L = load span in mm

(b) Master Template – A master template shall be produced to the same scale as the load vs. deflection data recorded from the toughness test. The master template shall be superimposed over the actual load deflection curves and compared with the recorded data.

(c) Acceptability Criterion – To conform to a given Toughness Performance Level the recorded data shall meet the specified flexural strength and fall above the line representing the given Toughness Performance Level at both the 0.5 mm and 2 mm net midspan deflection.

Table 209-C: Toughness Performance Level

Toughness Performance Level	Residual Flexural Strength By % of Design Flexural Strength	
	1/600 Span (0.5 mm)	1/150 Span (2.0 mm)
III	50%	30%

209.39.14 Boiled Absorption and Permeable Voids Test Specimens – Specimens for boiled absorption and permeable voids testing shall be 75 mm cubes cut from the broken ends of flexural test prisms, or extracted 75 mm diameter cores at least 100 mm long.

209.39.15 Boiled Absorption and Permeable Voids Testing – For each panel, three specimens shall be tested for boiled absorption and permeable voids at the specified age. Boiled absorption and permeable voids tests shall be conducted in accordance with ASTM C642. The report of results shall include all information recorded for "6.1 Calculation" in ASTM C642.

209.39.16 Laboratory Test Results – For each type of test, test results shall be considered acceptable if the average of three (3) specimens tested at a given age meets or exceeds the criteria specified in Table 209-A. The test results for an individual specimen shall not be less than 80% of the criteria specified for compressive strength, and not more than 120% of the criteria specified for boiled absorption and permeable voids.

209.40 Shotcrete Deficiencies and Repair

209.40.01 General – The Contractor shall, whenever possible, correct deficiencies while the shotcrete is still plastic. For any proposed changes to the initial mix design, a solid volume calculation shall be performed and submitted upon request. The calculation shall include the mass, density and volume of each constituent material, the total yield, and a comparison between measured (ASTM C138) and the calculated plastic density.

209.40.02 Deficiency Types – Shotcrete deficiencies other than failure to meet the performance requirements include, but are not limited to, the following:

(a) Improper consolidation of the shotcrete because of incorrect spraying angles or improper distance of the nozzle from the receiving surface;

(b) Improper control and removal of overspray and rebound material;

(c) Excessive shotcrete or fibre rebound;

(d) Any evidence of excessive plastic or drying shrinkage cracking;

(e) Segregation, incorporation of sand lenses, excessive voids, tears, sags, sloughs or delaminations in the work; and

(f) Inadequate thickness.

209.40.03 Shotcrete Repair – At the discretion of the Ministry Representative, shotcrete with deficiencies as defined above shall be removed and replaced to these specifications.

209.41 Shotcrete Evaluation – The Ministry Representative will implement a program of evaluation of the in-place shotcrete.

209.41.01 Core Extraction – When shotcrete is considered defective or deficient based on test results of test panels, the Ministry Representative may require cores to be extracted from the production shotcrete. Cores shall be extracted from the in-place shotcrete in accordance with CSA A23.2-14C, at locations detailed by the Ministry Representative, and tested for compliance to the project specifications. Generally, up to 6 cores shall be extracted from every 200 m² of shotcreted surface. Core extraction shall be at the Contractor's expense.

209.41.02 Non-Destructive Testing – The shotcrete will be checked for delaminations, voids or other deficiencies using appropriate non-destructive testing procedures.

MEASUREMENT

209.81 Shotcrete – Shotcrete will be measured by the CUBIC METRE supplied and installed in-place on the slope in the designated areas.

PAYMENT

209.91 Shotcrete – Payment for SHOTCRETE will be at the Contract Unit Price per cubic metre. Payment for up to 75% of the shotcrete installed will be authorized after the Ministry Representative has received 7 day test results for this shotcrete. Payment for the remaining 25% will be

authorized after 28 day test results and complete documentation have been received. The Ministry may consider reduced payment for defective or deficient shotcrete based on test results on test panels and/or in-situ shotcrete. The Contract Unit Price for shotcrete shall be considered full compensation for all requirements in this specification. The Ministry will not pay for volume losses to rebound, sloughing, rejected or otherwise discarded shotcrete.

APPENDIX 209-A
DAILY SHOTCRETE RECORD

File #: _____
PROJECT NAME_____

PROJECT NO._____

(TO BE SUBMITTED WITHIN 1 DAY AFTER EACH DAY'S OPERATION)

Contractor Name _____ Date _____
Nozzleman Name _____Weather _____
Helper Name _____ Temperature_____°C
Contractor's Consultant's Name _____ Contractor's Start _____End _____Time
Contractor's Consultant's Representative _____ Shotcrete Equipment Type_____

Shotcrete Type: (circle) WET DRY Process

For wet process, when is slump measured (circle)

BEFORE AFTER Air entrainment added
BEFORE AFTER Superplasticizer added
BEFORE AFTER Fibre Reinforcement added

On-Site Details and Tests for Shotcrete:

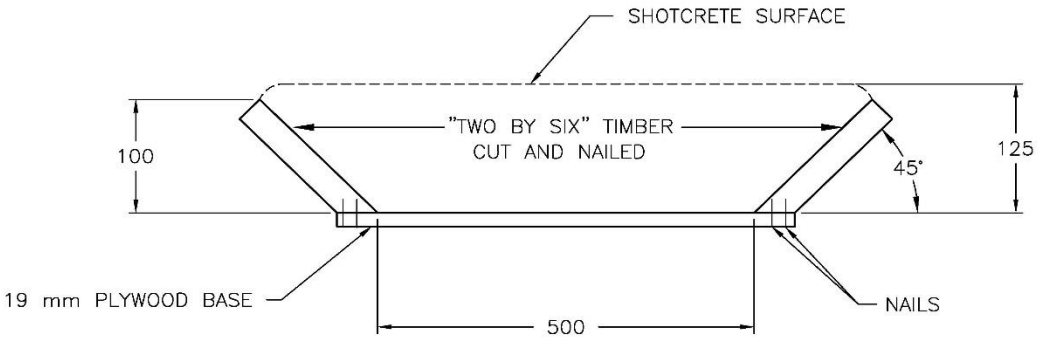
Batch #	Location		Mix Design #	Batch Times			Spray Time		Quantity Used (m^3)	Water/ Cement Ratio	Sample (As-Shot) Air Content (%)	Slump (wet mix) (mm)	Re-bound est. (%)	Average Thickness of In-Place Shotcrete (mm)	Test Panel No.
				From Plant	On-Site										
	Zone	Area			From	To	Start	End							

Comments: _____
_____
_____

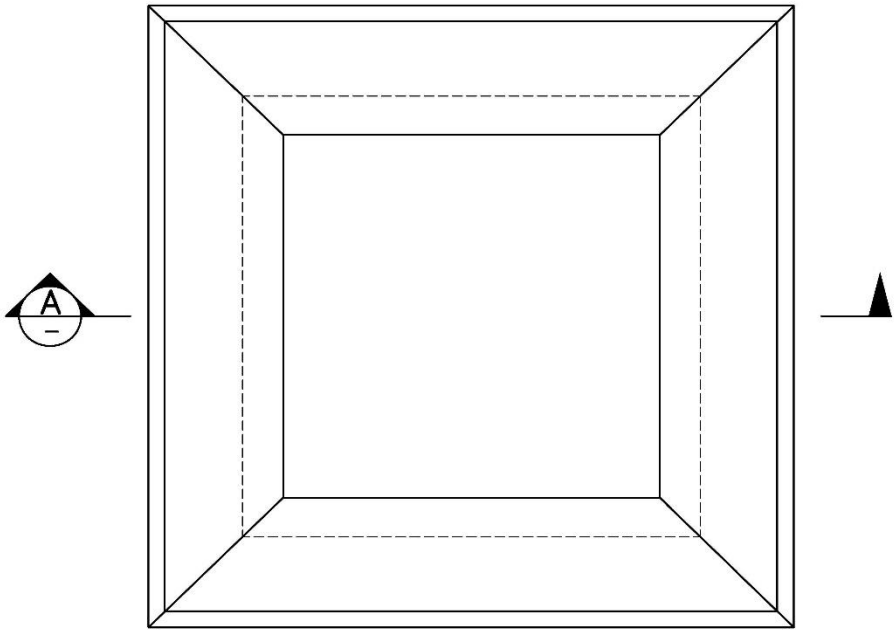
Certified Correct: _____ _____
(Contractor's Consultant's Representative) (Title)

(Use Additional Sheets if Necessary)

SHOTCRETE TEST PANEL FORM	SP209–01



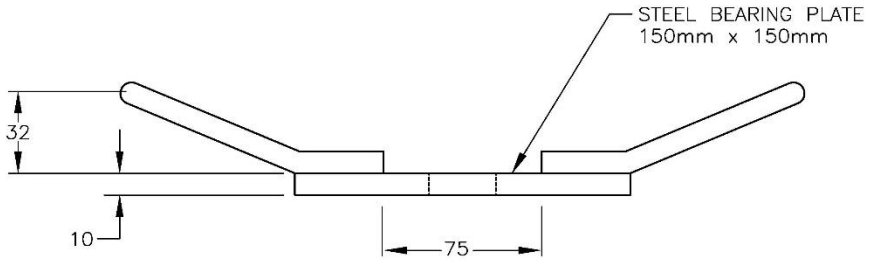
SECTION A–A



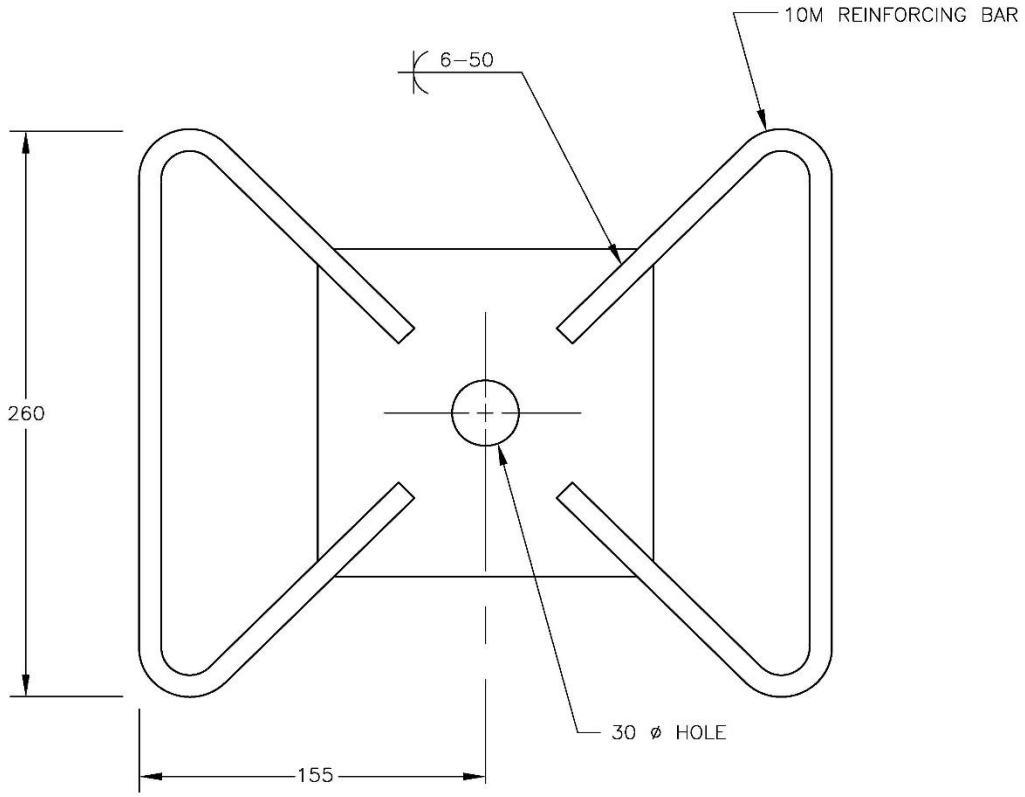
PLAN VIEW

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SHOTCRETE SPIDER PLATE DETAIL	SP209–02



ELEVATION VIEW



PLAN VIEW

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED